

Later scientists, including German biologist Theodor Boveri, had furthered this research, but the process behind inheritance remained unclear.

Early in his career, Morgan was skeptical about much of the previous research, including Mendel's theories of inheritance and Darwin's theory of natural selection. Naturally critical, and favoring controlled laboratory experiments over observational science, Morgan decided to carry out his own experiments to understand heredity.

The "Fly Room"

The subject Morgan chose for his experiments was *Drosophila*—fruit flies. They were ideal specimens: they displayed a wide range of physical traits, had only a small number of large chromosomes—which carry inheritable genetic information in the form of genes—and reproduce at great speed and in vast numbers. In 1908, Morgan set up a large laboratory at Columbia University, known as the "Fly Room," in which he

"We geneticists should rejoice, even with our noses to the grindstone."

Thomas Hunt Morgan, 1932

bred fruit flies by the millions. He cross-bred fruit flies with particular traits and analyzed the variations in their offspring by studying their chromosomes under a microscope.

Discovering inheritance

The key discovery came when a white-eyed male fly was bred with a red-eyed female. All the offspring had red eyes, indicating red was the dominant color while white was recessive. Morgan then cross-bred two of these offspring: one in four of the second-generation offspring had white eyes, and every

HERMANN JOSEPH MULLER

An eminent US geneticist, Hermann Joseph Muller received a Nobel Prize for his work on genetic mutations, which proved hereditary changes could be artificially induced through X-rays.



Brought up in New York, Muller (1890–1967) obtained a scholarship to Columbia University in 1907 to study biology and gained his PhD in 1916. He was inspired by the work of Thomas Hunt Morgan, and from 1920 spent 12 years at the University of Texas investigating genetic mutations. In 1926, he conducted a series of experiments that induced genetic mutations through the use of X-rays. This original and significant work secured his position as a renowned geneticist. Muller also became a pioneer in raising awareness of the long-term dangers of human exposure to radiation and campaigned for controls on nuclear weapons.

Morgan's experiments with fruit flies enabled him to trace the inherited physical characteristics across generations and determine the exact behavior of genes. The second generation of cross-bred flies revealed the white-eyed trait passed through the male line.